

Visual Studio interface showing the development of a C++ application named "Bank Account Simulation".

Code Editor (Bank Account Simulation.cpp):

```
28     fstream MyFile("transactions.txt", ios::app);
29     MyFile << "Withdrawal of $" << amount << " has failed. | " << "Balance: $" << balance << endl;
30     MyFile.close();
31     return;
32 }
33
34 balance -= amount;
35 cout << "You have withdrawn $" << amount << " and now have a balance of $" << balance << endl << endl;
36 fstream MyFile("transactions.txt", ios::app);
37 MyFile << "Withdrawn Amount: $" << amount << " | " << "New Balance: $" << balance << endl;
38 MyFile.close();
39
40 void purchase(string item, double cost, int quantity) {
41     double totalCost = cost * quantity;
42     if (totalCost > balance) {
43         cout << "This purchase of " << quantity << " " << item << " cannot be completed at this time. The total cost $" << totalCost << " exceeds you
44         fstream MyFile("transactions.txt", ios::app);
45         MyFile << "Failed purchase of " << quantity << " " << item << " for $" << totalCost << endl;
46         MyFile.close();
47         return;
48     }
49     balance -= totalCost;
50     cout << "You have purchased " << quantity << " " << item << " for $" << totalCost << endl << endl;
51     fstream MyFile("transactions.txt", ios::app);
52     MyFile << "Purchased " << quantity << " " << item << " for $" << totalCost << endl;
53     MyFile.close();
54 }
55
56 void transactionHistory() {
57     cout << "Here is your transaction history: " << endl << endl;
58     fstream MyFile("transactions.txt", ios::in);
59     if (!MyFile) {
60         cout << "You do not have access to this information. " << endl;
61         return;
62     }
63 }
```

Output Window:

```
Build started at 5:17 PM...
1>----- Build started: Project: Bank Account Simulation, Configuration: Debug x64 -----
1>Bank Account Simulation.cpp
1>Bank Account Simulation.vcxproj -> E:\VS\source\repos\Bank Account Simulation\x64\Debug\Bank Account Simulation.exe
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
===== Build completed at 5:17 PM and took 04.282 seconds =====
```

Solution Explorer: Shows the project structure for "Bank Account Simulation", including References, External Dependencies, Header Files, Resource Files, and Source Files.

Status Bar: 99 % No issues found. Ln: 46 Ch: 28 SPC CRLF.

Bottom Bar: Item(s) Saved. Add to Source Control. Select Repository.

6.5 E:\VS\source\repos\Bank Account Simulation\x64\Debug\Bank Account Simulation.exe

You cannot withdraw \$100, because you only have \$30

Here is your transaction history:

Deposited Amount: \$50 | New Balance: \$50.

Withdrawn Amount: \$20 | New Balance: \$30

Withdrawal of \$100 has failed. |

Balance: \$30

Purchased 1 Cereal for \$2.75.

Purchased 2 Cereal for \$5.5.

Failed purchase of 100 Cereal for \$275

Purchased 100 Cereal for \$275.

Deposited Amount: \$50 | New Balance: \$50

Withdrawn Amount: \$20 | New Balance: \$30

Withdrawal of \$100 has failed. | Balance: \$30

You have purchased 1 Cereal for \$2.75

You have purchased 2 Cereal for \$5.5

This purchase of 100 Cereal cannot be completed at this time. The total cost \$275 exceeds your balance of \$21.75

Your current balance is: \$21.75.

Would you like to see your transaction history? (yes or no)

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yes

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E:\VS\source\repos\Bank Account Simulation\x64\Debug\Bank Account Simulation.exe (process 23804) exited with code 0 (0x0).

Press any key to close this window . . .